

HAMZA BILAL

Full-Stack MERN Developer | AI / ML Engineer

Vehari, Punjab, Pakistan • +92 334 7348886 • hamzabilal.cs.pk@gmail.com
github.com/hamzabilal000 • linkedin.com/in/hamzabilalcs • fiverr.com/hamzabilal000

PROFESSIONAL SUMMARY

Full-stack MERN and AI/ML developer who ships production-grade systems end to end from MongoDB data models and Express APIs to React frontends and Python AI microservices. Sole architect of two complete platforms that integrate large language models (Claude, Gemini) into real-world products for Pakistan's agriculture and fintech sectors. Seeking full-time and freelance roles to build scalable, high-impact web and AI applications.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), Python, C/C++, Java, Bash

Frontend: React.js, Vite, React Router, Tailwind CSS, Recharts, HTML5 / CSS3, TypeScript

Backend: Node.js, Express.js, Python FastAPI, REST API design, JWT auth, bcrypt

Databases: MongoDB (Mongoose, Aggregation Pipeline, indexing, sharding), MySQL

AI / ML: Scikit-learn, TensorFlow / Keras, Pandas, NumPy, Decision Trees, K-Means, PCA

AI Integration: Claude & Gemini APIs (SSE streaming), Groq API, LangChain, VADER sentiment analysis

Tools: Git / GitHub, Postman, VS Code, Google Colab, Figma, npm

PROJECTS

GreenVault — Agri-Supply Chain ERP

2026

Final Year Project (COMSATS) • MERN + Python/Flask ML Microservice • Sole Developer

- Architected a production-grade MERN ERP for Pakistan's agri/pesticide supply chain with **14 MongoDB collections and 75k+ records**; JWT (cookie + Bearer) auth, StockLog/CashBook financial tracking, auto-generated sequential IDs, and a 6-hour alert scheduler.
- Built **"Crop Doctor AI"** — a diagnostic module pairing MongoDB aggregation with Claude API SSE streaming to identify crop diseases and recommend pesticides in real time.
- Authored an **IEEE 830-compliant SRS** and an end-to-end ML pipeline (Node/Express backend on port 5005, Python/Flask service on port 8000) single-handedly.

PSX AI Portfolio Platform

2025

Full-Stack Investment App • React + Express + Python FastAPI + Google Gemini

- Built an AI-driven platform for Pakistani retail investors: portfolio tracking, risk analysis, stock comparison, and a Gemini-powered chat assistant across **68 PSX stocks in 18 sectors**.
- Engineered an **8-factor recommendation engine** (P/E, dividend yield, ROE, beta, volatility, risk & goal alignment) plus a FastAPI service computing Sharpe ratio, volatility, and VaR from live yfinance prices.
- Implemented secure auth (JWT httpOnly cookies, email-OTP verification, password reset), RSS-based market news with VADER sentiment, and a themeable dark/light React UI.

ContentForge — AI Content Generator

2024

Personal / Freelance SaaS • React + Express + Groq API

- Built an AI SaaS generating SEO blog posts, social captions, and product copy on demand, with Tailwind UI, JWT-protected routes, and MongoDB-stored generation history.

EXPERIENCE

Freelance Full-Stack & ML Developer

2026 – Present

Fiverr — Self-Employed • Remote

- Deliver end-to-end MERN development and ML integration for global clients, focused on production-ready architecture and clean, well-documented codebases.

EDUCATION

B.Sc. in Computer Science (BSCS)

2023 – 2027 (expected)

COMSATS University Islamabad, Vehari Campus • CGPA: 3.69 / 4.0

- Relevant coursework:** Machine Learning, Advanced Database Systems, Advanced Web Technologies, Theory of Automata, Data Structures, Object-Oriented Programming.